

Forest Fire Detection System

Gurjeet Singh^{1*}, Abhishek Singh², Aman Kumar Patel², Ankita Paul², Haiqua Naaz²,
Md Araib Ashif², Nilava Bepari²

Abstract

The IOT industry is rapidly growing in the current scenario. Each one of us now understands the significance of Internet of Things. At present, it is being used in almost all industries making our lives a bit easier. Forest fires have become a massive threat across the globe, causing numerous negative impacts on human habitats and forest ecosystems. Climatic changes and the greenhouse effect are among the consequences of such destruction. Sadly, a higher percentage of forest fires occur due to human activities. Therefore, to minimize the catastrophe caused by forest fires, there is a need to detect them in their initial stage. This study proposes a system and methodology that may be used to detect forest fires at the initial phase using IoT. This study emphasizes detecting forest fires as early as possible by measuring various parameters which can lead to forest fires. An IoT-based forest fire detection system is suggested in this study to detect fires by monitoring values through sensors like the DHT22 temperature and Humidity sensor, MQ5 Smoke Sensor, Capacitive soil moisture sensor, and Flame sensor. The values are retrieved from the sensor through a Wi-Fi-enabled microcontroller (NodeMCU) and uploaded to the cloud. When these values reach a certain threshold, a message or a call is triggered and received by the user. This approach can potentially save lives or at least prompt the implementation of safety measures. The main aim of this study is to propose a system that can detect potential fire hazards at the earliest possible stage, thus preventing the occurrence of horrific wildfires. Our project has the potential to predict and detect fires, sending notifications to responsible authorities for immediate action before it is too late to control the forest fire.

Keywords: Forest fires, WSN, detection, visualization, alert system, fire prevention

INTRODUCTION

In India, the summer season has witnessed the highest number of forest fires, especially during the months of April to June. This period is attributed by low humidity, high temperature, dry vegetation, hence creating favorable conditions for the occurrence and spread of forest fires. According to the Forest Survey of India, a number of 52,785 forest fires were detected using MODIS (Moderate Resolution Imaging Spectro-radiometer) sensor and 3,45,989 forest fires were detected using SNPP-VIIRS (Soumi-National Polar-orbiting Partnership-Visible Infrared Imaging Radiometer Suite) in forest season from November 2020 to June 2021.

*Author for Correspondence

Gurjeet Singh
E-mail: gurjeetsingh714@gmail.com

¹Assistant Professor, Department of Electronic and Communication Engineering, Asansol Engineering college, Asansol West Bengal, India

²Student, Department of Electronic and Communication Engineering, Asansol Engineering college, Asansol West Bengal, India

Received Date: May 16, 2023

Accepted Date: June 17, 2023

Published Date: June 23, 2023

Citation: Gurjeet Singh, Abhishek Singh, Aman Kumar Patel, Ankita Paul, Haiqua Naaz, Md Araib Ashif, Nilava Bepari. Forest Fire Detection System. Journal of VLSI Design Tools & Technology. 2023; 13(1): 1–5p.

Forest fires pose huge threat to wildlife as well as economy. This can be in terms of loss of wildlife habitat, damage to timber, other forest resources as well as the tourism industry. One notable example of such forest fire which caused significant wildlife and economic losses in India was the 2016 fire in Uttarakhand, which kept on burning for several months. According to a report by the comptroller and Auditor General of India (CAG), the total monetary loss due to the forest fires was estimated at around 2,500 crore (approximately USD 340 million).

Overall, the collateral damage caused by these can be significant, hence alarming the need of effective forest fire management and prevention measures. This study focuses on development of smart network system which collects accurate data for the parameters related to forest fire. Wireless Sensor Network (WSN) is used where the collected data is uploaded to IoT cloud and continuous visualization of data is performed and an alert is triggered after the values reach certain predetermined threshold. This can prove to be a fast and cheap solution than obtaining satellite data. Also, the collected data can be used for further analysis to help reduce future occurrences of forest fires. Hence allowing prompt detection and immediate response in order to limit the fire's spread.

RELATED WORK

WSN comprises of various different sensor nodes put together spatially which are connected to a central node where all the data is received from all the different sensors. Each of the sensor nodes comprises of single or multiple sensing elements. Apart from sensors, there is a data processing unit, a power source (usually a battery), and a communication component. The sensed data is sent to the central node which further sends it to the desired end. WSNs are generally used for monitoring applications like border monitoring, environment monitoring, gathering meteorological variables, etc. One such application of WSN as elaborated by Kadir *et al.*, who placed several small nodes to detect the environmental changes and sent this data to the central cluster node which in turn transferred it to the server through the gate for easy maintenance and scalability of data [1].

Another method for monitoring and detection of forest fire as discussed by Liu *et al.* uses data aggregation in WSNs [2]. This paper proposes a method which provides faster and a bit more effective reaction to forest fires by consuming validated WSN's energy, therefore maximizing the network's operational efficiency. This is also confirmed by large number of experiments in simulation. WSN can be a better choice for the tasks like detection of earthquake, landslide, flood, forest fire, etc. [3]. To minimize the power consumption of sensors, techniques like keeping only selective components active while rest are deactivated were proposed [4–6].

Several other researches on WSNs were carried out. One of which proposed fusion of information from multiple sources which in turn was considered in final decision making, which actually is better than using sources independently, after fusion two algorithms based on the threshold ratio and Dempster-Shafer theory were used [7].

Furthermore, research conducted by Singh *et al.*, in order to enhance the detection accuracy proposed application of machine learning approaches, such as support vector machine (SVM) classification and regression techniques such as logistic regression [8]. Although this approach poses limitations like limited energy, energy required to process the data, limited computations, and complex ML algorithms [9, 10].

Availability of power can be a challenging aspect in forest areas; relying solely on battery power can be difficult as they do not last long, and power distribution via wire would be costly. Hence researchers proposed the idea of using solar powered systems as secondary power source along with rechargeable batteries as the primary power source [7, 11].

PROPOSED WSN SYSTEM

In most of the cases, cause of forest fires can be both natural and man-made, throughout the globe. Particularly in summer season, the environment becomes hot and dry leading to a favorable condition for the occurrence of forest fires, therefore, acting as a fuel for them. Another significant cause of such disasters is global warming. Each such event not only destroys thousands of hectares of forest land, but also has a huge impact on wildlife habitat, economy, resources and facilities and public assets. Apart from these, firefighters and locals face high risk of horrific accidents each year.

Hence detection of such forest fires in early stages becomes necessary in order to avoid such horrific consequences. In this study, we have proposed a model that consists of, array of wireless sensor networks which keep on monitoring various parameters related to forest fire, such as temperature, humidity, soil moisture and concentration of smoke in a particular region (Figure 1 block diagram). Simultaneously the data from all the sensor nodes, is sent to a central node (NodeMCU), which uploads the collected data to an IoT based cloud (ThingSpeak).

On the other hand, the data is continuously visualized on the cloud using graphs and widgets. There is a predetermined threshold for each sensor, once it is reached, an alert is triggered using a web-based service-IFTTT (If This, Then That) which connects different online platforms and services via applets and sends alert via email, message, automated call, etc.

Hardware that can be used to monitor and detect such events are available in various kinds. Our system uses a flame sensor, MQ5 smoke sensor, capacitive soil moisture sensor, DHT22 humidity and temperature sensor. All these sensors send their data to the base station, which then uploads it on the server for further visualization, analysis and decision making.

RESULT AND DISCUSSION

The proposed design of forest fire detection system has been tested thoroughly. The sensors used here are able to detect all the environmental parameters, which pose potential threat to forest areas, with desired accuracy. The design mainly comprises of four sensors: DHT22 sensor, MQ5 sensor, capacitive soil moisture sensor, and flame sensor. Figure 2 shows the above-mentioned model. The system was tested in various environmental conditions. Sensor readings for different environmental parameters have been tested extensively and the data got logged successfully on the cloud based IoT monitoring system.

Figure 3 shows graph for different parameters recorded by the sensors for a particular period of time. Also, the call and message-based alert were tested successfully after simulating an occurrence of fire using a controlled candle flame. Similarly, smoke from burnt paper was used for testing the smoke sensor, dry soil for soil moisture, and hot air blower for temperature and humidity. Changes in the value of parameters occurred as expected and their respective alerts were triggered successfully on the user's device.

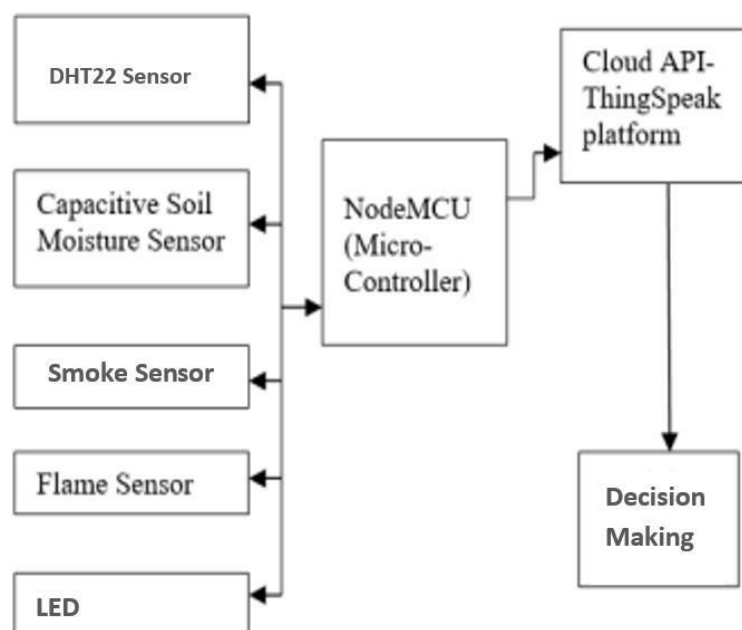


Figure 1. Block Diagram of the proposed system.

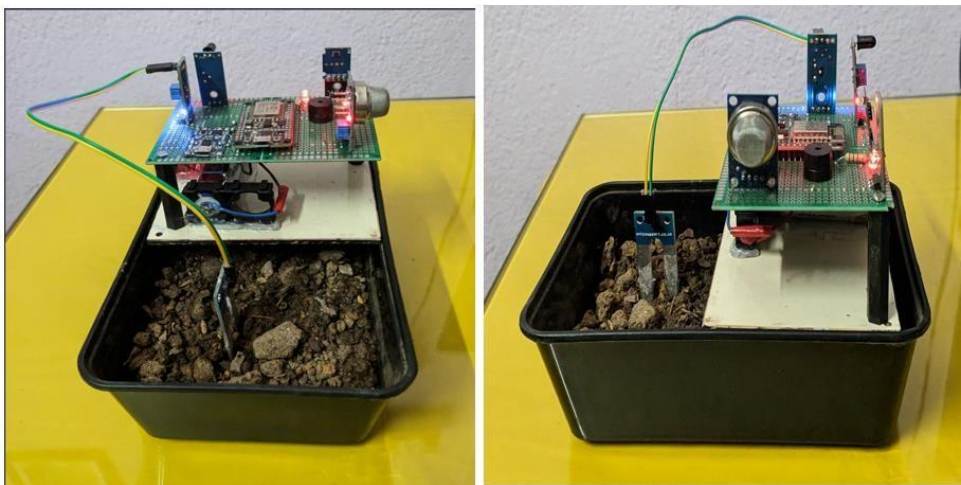


Figure 2. Sensor node assembled for fire detection.



Figure 3. Test results for a particular period of time.

CONCLUSION

It has been proposed to develop and enhance the already existing WSN node to monitor and detect forest fires in early stage for accurate detection. Various basic environmental parameters which directly have an impact on forest fires were successfully detected by the sensor setup, also some additional parameters of forest fire were taken into consideration. The data visualization of parameters as shown in graph were accurate and responsive. Further improvement and enhancement of the sensing node via various other techniques like camera imaging, intelligent programming algorithm, is one of the targets for the future work in this research. Finally, at the end of the project, the highly applicable WSN system proposed to detect the forest fires in initial stage turns out to be very useful for the prevention of forest fires.

REFERENCES

1. Kadir EA, Irie H, Rosa SL. Modelling of wireless sensor networks for detection land and forest fire hotspot. International Conference on Electronics, Information, and Communication (ICEIC). 2019; 1–5.
2. Liu Y, Liu Y, Xu H, Teo KL. Forest fire monitoring, detection and decision-making systems by wireless sensor network. Chinese Control and Decision Conference (CCDC). 2018; 5482–5486. doi:10.1109/UKSim.2013.133
3. Diwaker Pant, Sandeep Verma, Piyush Dhuliya. A study on disaster detection and management using WSN in Himalayan region of Uttarakhand. 2017 3rd International conference on advances in computing, communication & automation (ICACCA) (Fall). 2017; 1–6.
4. Bayo A, Antolín D, Medrano N, Calvo B, Celma S. Early detection and monitoring of forest fire with a wireless sensor network system. Procedia Eng. 2010; 5: 248–251.
5. Alkhatib AA. Wireless sensor network for forest fire detection and decision making. International Journal of Advances in Engineering Science and Technology (IJAEST). 2013; 2(3): 299–309.
6. Sakib Abdullah, Sandor Bertalan, Stanislav Masar, Adem Coskun, Izzet Kale. A Wireless Sensor Network for Early Forest Fire Detection and Monitoring as a Decision Factor in the Context of a Complex Integrated Emergency Response System. 2017 IEEE Workshop on Environmental, Energy, and Structural Monitoring Systems (EESMS). 2017; 1–5. doi: 10.1109/EESMS.2017.8052688.
7. Díaz-Ramírez A, Tafoya LA, Atempa JA, Mejía-Alvarez P. Wireless Sensor Networks and Fusion Information Methods for Forest Fire Detection. Elsevier Ltd., Procedia Technol. 2012; 3: 69–79. doi:10.1016/j.protcy.2012.03.008.
8. Singh Y, Saha S, Chugh U, Gupta C. Distributed event detection in wireless sensor networks for forest fires. In 2013 UKSim 15th International Conference on Computer Modelling and Simulation, Cambridge. 2013; 634–639.
9. Kansal A, Singh Y, Kumar N, Mohindru V. Detection of forest fires using machine learning technique: A perspective. In 3rd International Conference on Image Information Processing, India. 2015; 241–245. doi:10.1109/ICIIP.2015.7414773.
10. Zhang T, Zhao Q, Nakamoto Y. Faulty sensor data detection in wireless sensor networks using logistical regression. In 2017 IEEE 37th International Conference on Distributed Computing Systems Workshops (ICDCSW), Atlanta. 2017; 13–18. doi: 10.1109/ICDCSW.2017.37.
11. Sakr George E, Ajour Rafik, Khaddaj Areej, Saab Bahaa, Salman Alaa, Helal Ola, Elhadj Imad H, Mitri George. Forest fire detection wireless sensor node. Advances in Forest Fire Research. Imprensa da Universidade de Coimbra; 2014; 1395–1406. doi:10.14195/978-989-26-0884-6_153.